

Evidence to answer

Server-side request flow / PS3 / 06 OCT 2026

1 / Browser + ChatGPT sign-in

Question only; no API key or chosen role. Registration follows authenticated identity. Public application pages read /api/data; hosted audience controls access.



2 / Sites backend checks access

Trusted identity header -> D1 users -> registration/role check -> same-origin check -> atomic rate limit. Anonymous and unregistered adviser calls are rejected.



3 / D1 supplies current evidence

Prepared queries read saved model inputs, 19 claims, metrics and source records. First-phase engineering assumptions are persisted. Missing values remain NULL.



4 / OpenAI + controlled tools

Server-only secret; current D1 evidence is untrusted data, not instructions. Tools read design/country data or calculate energy/scenarios; at most three model turns.



5 / Approved external API

World Bank fixed historical endpoint. Editor/admin refresh validates country, indicator, year, unit, range and complete response before an atomic D1 batch. Failure retains data.



6 / Validate -> audit -> browser

Answer schema restricts citation IDs to real sources. Server checks output, records token totals, and returns answer, source links, assumptions and uncertainty. No question text is audited.

Storage: users, countries, sources, metrics, designs, design_claims, events, rate_limits. Schema migrations and separate idempotent seeds preserve saved data.

Verification boundary: real model tests cover saved-input changes, missing facts, source citations, certification refusal and one malicious-source fixture. These are finite tests, not a general safety or engineering guarantee. Teacher access remains a separate submission step.